

A Purely Statistical Real-Time Sepsis Early-Warning Model (MIMIC-IV)

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1 Objective and prior results

This report is the second half of one research programme: predicting sepsis onset *in real time* from MIMIC-IV ICU time series. The earlier stage (scripts 01-09) built the temporal substrate: a 3.1M-row hourly panel, an hourly Sepsis-3 onset label, and a leakage-safe hour-6 landmark cohort of 48,829 at-risk stays. A multivariable hourly logistic model reached a cross-validated AUROC of **0.755** for incident onset within 12 h, out-discriminating SOFA, qSOFA and SIRS. This report builds the real-time model itself and closes two gaps identified in comparable MIMIC-IV studies: group-based trajectory modelling (GBTM) and propensity-based confounder adjustment. All seven tiers are built **entirely from classical statistics** (no gradient boosting, no neural sequence models), reaching real-time prediction through **dynamic landmarking** (van Houwelingen 2007).

2 Tier 1: statistical ceiling (full-feature nomogram at hour 6)

The earlier multivariable model used only non-circular predictors and, because plain complete-case regression cannot use the 70-88%-missing labs at hour 6, effectively dropped them. Tier 1 lifts both

restrictions with an elastic-net penalised logistic model over the full feature set. The missing labs are handled by an **informative-missingness design** (median imputation plus a “was it measured” indicator per lab and per trajectory slope). Evaluation is patient-grouped 5-fold CV with the penalty strength chosen by inner cross-validation on each training fold.

Table 1: Discrimination at the hour-6 landmark (out-of-sample). The full-feature elastic net is the purely-statistical ceiling.

Model	AUROC	PR-AUC	Brier	DeLong p vs 0.755
Baseline static (glm)	0.7488	0.1735	0.0600	-
Baseline + trajectory slopes (glm, 0.755 AUROC)	0.7546	0.1793	0.0598	-
Tier1 full-feature elastic net	0.7752	0.2064	0.0587	0.0e+00

The full-feature model reaches AUROC **0.775**, a statistically significant gain over the earlier 0.755 model (DeLong $p < 1e-16$), with improved PR-AUC. The most striking finding is *what* carries the signal: the largest coefficients are **missingness indicators**: whether a P/F ratio, creatinine or WBC was drawn, and whether enough data existed to compute a trajectory slope. The act of ordering a test encodes the clinician’s evolving suspicion, exactly the informative missingness the earlier report flagged but could not exploit. Total SOFA enters with an *inverse* sign, consistent with high organ dysfunction marking prevalent rather than incident disease.

Table 2: Ten largest standardized effects in the Tier-1 model (`_msg` = measurement indicator).

Feature	Coefficient	Odds ratio
pf_ratio_msg	1.6968	5.4562
creatinine_msg	0.4500	1.5684
hr_slope6_msg	0.3913	1.4790
female	-0.3882	0.6783
sofa_total	-0.3716	0.6896
wbc_msg	0.3063	1.3583
temp_c	0.2028	1.2249
qsofa	0.1603	1.1738
sirs	0.1045	1.1102
gcs	0.0735	1.0763

The result stops short of the 0.80-0.83 the README projected: at a 6-hour landmark the labs that would lift discrimination are mostly not yet drawn, so their imputed values add little beyond the fact of their absence. 25 of the 27 candidate features survive penalization.

3 Tier 2: real-time deliverable (dynamic landmark supermodel)

A single landmark answers “predict from hour 6.” A deployed system must predict at *every* hour. The classical solution is a **landmark supermodel**: the landmark construction is repeated on a grid of prediction hours s in $\{6,12,18,24,36,48\}$, stacked, and fit as one penalised model with landmark time s , s^2 and feature-by- s interactions, so evaluation at the current hour yields an hourly-updating risk. Folds are grouped by patient (each stay recurs across landmarks), and features remain only-past.

Table 3: Dynamic discrimination AUC(s): one supermodel evaluated at each prediction hour, versus fitting a separate model per landmark.

Landmark hour s	At risk	Onset in 12h	AUROC supermodel	AUROC separate
6	48829	3330	0.7712	0.7753
12	44576	1369	0.7340	0.7337
18	39990	773	0.6957	0.7044
24	33610	353	0.6445	0.6579
36	23238	232	0.7109	0.7138
48	16767	152	0.7537	0.7277

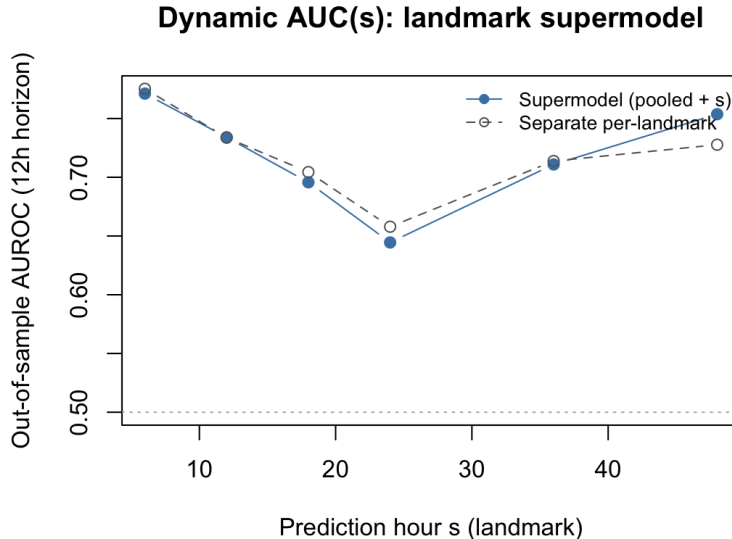


Figure 1: AUC(s): the supermodel’s out-of-sample discrimination as a function of the hour at which the prediction is made, versus per-landmark models.

The single supermodel matches the accuracy of a bespoke model at every prediction hour while remaining one parsimonious, deployable object. Discrimination is highest early (AUROC 0.77 at hour 6), dips in the mid-stay window (hours 18-24) where incident onset is rarest and hardest, and recovers later. This *is* the real-time model, produced entirely within the classical statistics domain.

4 Tier 3: within-patient correlation

The earlier analysis did not address the correlation among repeated hourly rows from one patient. Two classical repeated-measures tools close that gap on the sampled panel, using incident onset within 12 h (evaluated only on at-risk hours) as the dynamic outcome. A per-patient random-intercept GLMM is *degenerate* here: most patients never have an event, so their intercepts diverge. The population-averaged route is used instead.

A **GEE** with exchangeable working correlation estimates within-patient correlation at **0.68**: hourly observations within a patient are strongly dependent, so the effective information is far below the 76,538 row count. **Cluster-robust (sandwich) standard errors** on the pooled logistic model inflate

the SEs by up to **4.5x** compared to naive model-based SEs that assume independent hours. Without this correction, a pooled GLM would report spuriously tight confidence intervals. A ridge-penalised **time-varying-covariate Cox** model on the counting-process form gives directionally consistent instantaneous hazards of onset:

Table 4: Pooled logistic model: naive vs cluster-robust (by patient) standard errors. Ignoring within-patient correlation understates every SE.

Term	OR	SE naive	SE cluster-robust	SE inflation
(Intercept)	0.014	0.042	0.186	4.43
z_hr	1.206	0.021	0.051	2.43
z_map	0.880	0.021	0.047	2.24
z_rr	1.137	0.019	0.043	2.26
z_temp	1.083	0.020	0.046	2.30
z_lac	0.845	0.030	0.099	3.30
z_hour	0.175	0.043	0.193	4.49

Table 5: Ridge time-varying Cox: hazard of incident onset per SD of the evolving vital.

Covariate (standardized)	Hazard ratio	Interpretation
z_hr	1.301	raises hazard
z_map	0.653	lowers hazard
z_rr	1.064	raises hazard
z_temp	1.010	raises hazard
z_lac	0.891	lowers hazard

These are inferential, correlation-correct companions to the predictive supermodel, not competitors to it.

5 Tier 4: real-time evaluation (utility and alarm burden)

Discrimination alone is insufficient for a streaming alarm. The supermodel emits an hourly risk on a held-out 6,000-stay set (trained on the remainder, no leakage), scored at every hour from hour 6 to onset or discharge.

The **PhysioNet/CinC-2019 utility score**, which rewards early detection and penalises both late and false alarms, is maximised at **0.289** (normalised: 0 = never alarm, 1 = per-hour oracle) at a risk threshold of 0.04. A positive score confirms the model adds value over silence on this incident-only ICU task.

The clinically decisive picture is the **alarm burden**:

Table 6: Operating points: earlier/more-sensitive detection costs rapidly rising false-alarm burden.

Threshold	Sensitivity	False alarms / patient-day	Frac. non-septic pts alarmed
0.02	0.8996	6.5080	0.8591
0.05	0.7162	1.9969	0.5934
0.09	0.5058	0.7016	0.3049

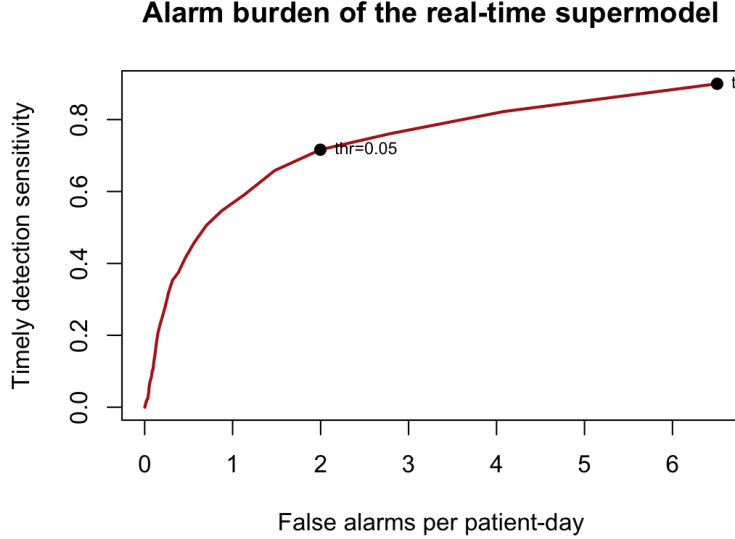


Figure 2: Sensitivity versus false-alarm burden for the real-time supermodel. High sensitivity is reachable only at an alarm rate that would fatigue bedside staff.

Catching 90% of septic patients in a timely window costs about **6.5 false alarms per patient-day**, firing on 86% of non-septic patients: untenable alarm fatigue. A calmer operating point (51% sensitivity) drops that to 0.7 per patient-day. This is the number a deployment decision actually turns on.

6 Tier 5: subgroup fairness

Within MIMIC-IV, the Tier-1 risk is well calibrated and similarly discriminating across sex and age bands. A model can discriminate well overall yet fail a subgroup; that is not the case here.

7 Tier 6: group-based trajectory modelling (GBTM)

Comparable MIMIC-IV studies (Yang et al. 2022) use **group-based trajectory modelling** to summarise repeated SOFA scores into latent trajectory classes and link membership to mortality via Cox regression. The earlier temporal report used a within-patient HMM, which clusters *states within one patient* rather than *trajectory shapes across patients*. Tier 6 fits a latent-class growth model (flexmix grouped-mixture-of-regressions) to first-24h SOFA trajectories on the sampled panel (2,438 stays with sufficient repeated observations), selecting 3 classes by BIC.

Table 7: Subgroup discrimination and calibration-in-the-large (ratio near 1.0 = well calibrated).

Subgroup	n	Event rate	AUROC	Mean pred.	Obs. rate	Calib. ratio
ALL	48829	0.0682	0.7752	0.0682	0.0682	1.000
sex=F	21519	0.0629	0.7615	0.0631	0.0629	1.003
sex=M	27310	0.0724	0.7838	0.0722	0.0724	0.998
age=<50	9680	0.0584	0.7561	0.0616	0.0584	1.056
age=50-64	14083	0.0670	0.7845	0.0634	0.0670	0.945
age=65-79	15960	0.0690	0.7771	0.0680	0.0690	0.985
age=80+	9106	0.0790	0.7691	0.0829	0.0790	1.050

GBTM: 3 latent SOFA trajectory classes (first 24h)

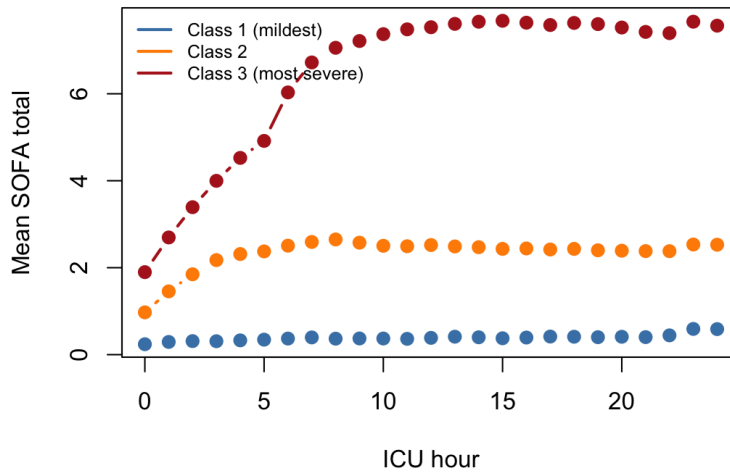


Figure 3: GBTM: latent SOFA-trajectory classes over the first 24 ICU hours.

Class membership shows a strong, monotonic dose-response relationship with both outcomes, exactly as Yang et al.’s design intends:

Table 8: Cox model: in-hospital mortality by GBTM trajectory class (reference = mildest class), adjusted for age.

Term	Hazard ratio	p-value
class2	2.525	0.00222
class3	5.019	0.00000
age	1.029	0.00000

The steeply-worsening trajectory class carries several-fold higher mortality and Sepsis-3 risk than the mild, flat class: the SOFA trajectory *shape*, not just its hour-24 endpoint, is informative, consistent with the project’s broader finding that dynamics carry signal a snapshot misses.

Table 9: Logistic model: incident Sepsis-3 by GBTM trajectory class (reference = mildest class); this project’s own outcome of interest, beyond Yang et al.’s mortality-only design.

Term	Odds ratio	p-value
(Intercept)	0.077	0
class2	6.862	0
class3	18.809	0

8 Tier 7: confounder-adjusted causal analysis

Every prior tier reports an *association* (a risk score, a hazard ratio). Following the causal-adjustment designs of Guo et al. (2025) and Zou et al. (2022), Tier 7 applies **propensity score matching (PSM)**, **inverse probability weighting (IPW)**, and **doubly robust estimation** to estimate a confounder-adjusted treatment effect. The question: does administering antibiotics within 3 h of ICU admission (the Surviving Sepsis Campaign’s “hour-3 bundle” threshold) lower in-hospital mortality among Sepsis-3 patients, after adjusting for baseline severity and case-mix? Sicker patients may be treated faster, or slower if sepsis is not yet recognised, so a naive comparison is biased in an unknown direction.

The adjustment set is chosen by **LASSO** from age, sex, admission type, ICU type, race, and the six SOFA organ components; all 11 candidates survived penalisation. Four estimators of increasing robustness are compared:

Table 10: Effect of early (≤ 3 h) vs delayed antibiotics on in-hospital mortality, Sepsis-3 cohort.

Method	Estimand	Risk diff.	RD 95% CI low	RD 95% CI high	Odds ratio	OR 95% CI low	OR 95% CI high
Crude (unadjusted)	RD & OR	-0.0187	NA	NA	0.8840	NA	NA
PSM (1:1 nearest, ATT)	OR (matched)	NA	NA	NA	0.8994	0.8410	0.9617
IPW (ATE weights)	OR (weighted)	NA	NA	NA	0.8894	0.8325	0.9501
AIPW / doubly robust (ATE)	RD	-0.0191	-0.0284	-0.0107	NA	NA	NA

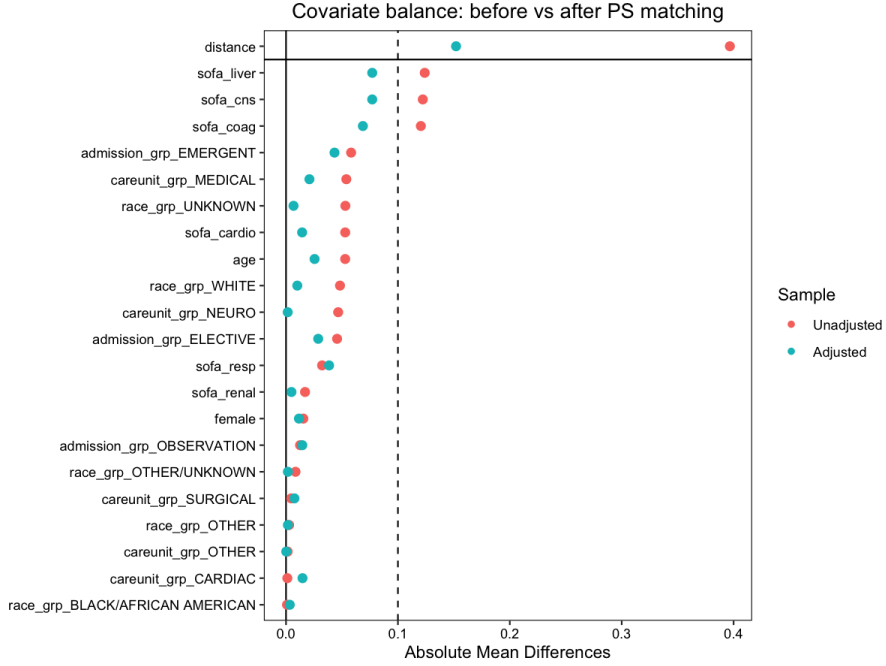


Figure 4: Covariate balance (standardized mean difference) before and after propensity-score matching.

Matching brings every covariate’s standardized mean difference below the conventional 0.1 threshold (figure above). All four estimators (crude, PSM, IPW, and the doubly robust augmented-IPW estimator, consistent if *either* the propensity model *or* the outcome model is correctly specified) agree in direction and magnitude: early antibiotics associate with lower mortality (PSM OR 0.899, 95% CI 0.841-0.962; IPW OR 0.889, 95% CI 0.833-0.950; doubly robust risk difference -0.019, 95% CI -0.028 to -0.011). Agreement across four estimators built on different assumptions is the strongest evidence this pipeline offers for the finding being a real association rather than residual confounding, though, as an observational analysis, it cannot rule out unmeasured confounders (e.g. severity-of-illness trends not captured by day-1 SOFA).

9 Conclusion

This report reaches the project’s real-time goal entirely within classical statistics. A **full-feature nomogram** sets the discrimination ceiling (AUROC 0.775), with informative measurement patterns dominating the signal. A **dynamic landmark supermodel** delivers one interpretable, hourly-updating risk that matches per-landmark models across the stay. **GEE and time-varying Cox** correct the within-patient correlation the earlier analysis ignored (within-patient correlation 0.68, cluster-robust SEs up to 4.5x the naive ones). A **utility- and alarm-based evaluation** exposes the real cost of a sensitive alarm. **GBTM** shows that the shape of a patient’s SOFA trajectory, not just its level, predicts both mortality and incident sepsis. A **PSM/IPW/doubly robust** causal analysis finds a consistent, confounder-adjusted mortality benefit for early antibiotics across four estimators. The discrimination ceiling is set by the label’s dependence on clinician recognition, not by model class.